A Multimodal, Explainable Hierarchical Framework for Skin-Lesion Classification with Diffusion-Based Augmentation of Rare Malignancies

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Abstract

Automated skin-lesion triage is usually framed as image-only classification, yet dermatologists routinely combine the dermatoscopic image with patient context — age, sex, and where on the body the lesion sits — all of which carry genuine diagnostic signal. We revisit a three-stage hierarchical classifier (benign vs. malignant → benign subtype → malignant subtype) built on EfficientNetB3 and upgrade it along four axes that matter for a clinical decision-support tool: (i) **multimodal late fusion** of image and patient metadata at every stage; (ii) **principled class-imbalance handling** using the full dataset with balanced class weights and on-the-fly augmentation, replacing the undersampling of the original pipeline; (iii) a **time-boxed diffusion augmentation study** in which a DDPM synthesises additional examples of the rarest malignant class (`akiec`) and we measure the effect on minority-class recall; and (iv) **medical-grade evaluation** — sensitivity/specificity, ROC-AUC, calibration (ECE + reliability diagrams), confusion matrices, Grad-CAM explanations, and **McNemar's test** for statistical significance between the image-only baseline and the multimodal model on an identical held-out set. All models are trained on HAM10000 with a single global stratified split shared across every stage, so the end-to-end pipeline is evaluated on images no sub-stage ever saw. On the held-out test set, the multimodal Stage-1 model reaches a malignancy **sensitivity of 53.6%** (baseline 47.8%) at 89.8% specificity, ROC-AUC 0.852; McNemar's test gives $p = 0.010$. The diffusion arm changes `akiec` recall from 18.4% to 26.5%. We report every result, including the stages where fusion does **not** help — a negative result is still a finding — and package the work as runnable Colab notebooks, an importable `src/` library with unit tests, and an explainable Streamlit demo.

1. Introduction

Skin cancer is among the most common malignancies worldwide, and melanoma in particular is highly curable when caught early and lethal when missed. This asymmetry — a missed melanoma (false negative) is far costlier than a false alarm (false positive) — should shape both the model and the way it is evaluated. Deep convolutional networks now match dermatologists on curated benchmarks [Esteva 2017; Tschandl 2019], but a model that only ingests pixels discards information a clinician always has: the patient's age, sex, and the lesion's anatomical site are established risk modifiers, and dermatoscopic "metadata + image" reasoning is standard clinical practice.

The starting point for this work was a competent but conventional pipeline: a three-stage hierarchy of EfficientNetB3 CNNs — Stage 1 separates benign from malignant, Stage 2a classifies benign subtypes {`nv`, `bkl`, `vasc`}, and Stage 2b classifies malignant subtypes {`mel`, `bcc`, `akiec`} — trained on class-balanced data obtained by **undersampling** the majority classes, and evaluated with **accuracy only**. That design has four weaknesses a reviewer will immediately flag, and this report is organised around fixing them:

1. **It is image-only.** It ignores the patient metadata that ships with HAM10000 and that clinicians use every day.
2. **Undersampling throws away data.** Balancing `nv` (≈ 6705 images) down to the size of the rare classes discards the majority of the training signal and inflates variance.
3. **Accuracy is the wrong headline metric** for an imbalanced screening problem. Sensitivity, specificity, ROC-AUC, per-class recall, and calibration are what matter, and none were reported.
4. **There is no explainability and no baseline comparison.** Nothing shows the network attends to the lesion rather than to hair, ink, or ruler marks, and nothing isolates *why* any design choice helps.

Contributions

- **A multimodal hierarchical classifier** with EfficientNetB3 image branch and an MLP metadata branch fused by concatenation before each stage's softmax (§4.2), trained end-to-end per stage.
- **An imbalance study** contrasting the original undersampling regime with full-data training under balanced class weights and live augmentation, and a **diffusion-augmentation ablation** on the rarest malignant class (§4.4, §6.3).
- **A Grad-CAM explainability layer** (§4.5) that overlays the Stage-1 evidence map on the lesion, surfaced directly in the demo app.
- **A rigorous, clinically-oriented evaluation protocol** (§4.6) with sensitivity/specificity, ROC-AUC (binary and one-vs-rest), ECE + reliability, confusion matrices, and **McNemar's test** for image-only vs. multimodal significance on a shared, leakage-free split.
- **A reproducible artifact**: six Colab notebooks, an importable and unit-tested `src/` package, a fill-from-JSON report pipeline, and an explainable demo.

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2. Related Work

Dermatological image classification. Esteva et al. [2017] demonstrated dermatologist-level classification with a single CNN, and the ISIC/HAM10000 benchmarks [Tschandl 2018; Codella 2018; Combalia 2019] standardised evaluation. EfficientNet [Tan & Le 2019] is a strong, parameter-efficient backbone widely adopted for this task; we keep the EfficientNetB3 backbone of the original project for a fair comparison.

Multimodal fusion of image and metadata. Combining dermatoscopic images with clinical metadata improves diagnosis both for clinicians [Haenssle 2018] and for models. Yap et al. [2018] and Kawahara et al. [2019] fuse image features with patient/lesion metadata and report gains over image-only baselines, motivating our late-fusion design. Our contribution here is not a new fusion operator but a controlled, significance-tested comparison at every stage of a deployed hierarchy, including the stages where fusion does not help.

Class imbalance and augmentation. HAM10000 is dominated by melanocytic nevi; naive training over- or under-fits accordingly. Standard responses are class weighting, oversampling, and photometric/geometric augmentation [Perez 2019]. More recently, **generative** augmentation with diffusion models [Ho 2020; Rombach 2022] has been applied to dermatology [Akrout 2023], with mixed but promising results for rare classes. We include a compact, resource-bounded DDPM arm precisely to test — rather than assume — that synthetic minority samples help.

Explainability and calibration. Grad-CAM [Selvaraju 2017] is the de-facto saliency method for CNNs and is important in medical imaging for detecting shortcut learning (e.g. attending to rulers or ink). Calibration [Guo 2017] — whether a "90% malignant" output is right 90% of the time — is essential if a clinician is to trust the

confidence, so we report ECE and reliability diagrams alongside discrimination metrics. For model comparison we use **McNemar's test** [McNemar 1947; Dietterich 1998], the standard paired test for two classifiers on one test set.

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3. Data & Preprocessing

Dataset. We use **HAM10000** [Tschandl 2018] (Kaggle `kmader/skin-cancer-mnist-ham10000`): 10,015 dermatoscopic images across seven diagnostic categories, shipped with `HAM10000_metadata.csv` (`image_id, dx, dx_type, age, sex, localization`). The seven `dx` codes map to our clinical groups as:

Group	`dx` codes	Meaning
Malignant	`mel`, `bcc`, `akiec`	melanoma, basal-cell carcinoma, actinic keratosis / intraepithelial carcinoma
Benign	`nv`, `bkl`, `vasc`, `df`	melanocytic nevi, benign keratosis, vascular, dermatofibroma

Stage 1 (benign vs. malignant) uses **all seven** classes. Stage 2a classifies the three benign subtypes {`nv`, `bkl`, `vasc`}; Stage 2b classifies the three malignant subtypes {`mel`, `bcc`, `akiec`}. `df` (dermatofibroma, ≈ 115 images) is grouped as benign at Stage 1 but is not sub-classified — too few examples for a reliable fourth benign head; this is a documented limitation (§9).

Leakage-free global split. A single stratified partition (70% train / 15% val / 15% test, seed 2024, stratified on the 7-way `dx`) is assigned **once** in `src/data.assign_global_split` and written to `models/ham_split.csv`. Every stage then derives its own train/val/test rows by *filtering* this shared column. This is the key design decision that makes the end-to-end evaluation honest: because the partition is global, an image used to train any sub-stage can never surface in the pipeline's held-out test set. Splitting each stage independently — the easy mistake — would leak Stage-2 training images into the Stage-1 test set and vice-versa.

Image preprocessing. Images are decoded (JPEG for real images, PNG for synthetic ones — a single `tf.io.decode_image` path handles both) and resized to 224 × 224. Pixels are kept in `[0, 255]`: EfficientNet carries its own `preprocess_input` normalisation internally, so an extra rescale would be a silent bug. Training augmentation (horizontal/vertical flip, ±15° rotation, zoom, brightness, contrast) is applied live in the `tf.data` pipeline and is value-range-aware so it too stays in `[0, 255]`.

Metadata preprocessing (`src/metadata.MetadataProcessor`). `age` is median-imputed (missing ≈ 1%) then z-scored; `sex` and `localization` are one-hot encoded with an explicit `unknown` bucket for missing or unseen values. The processor is **fit on the training split only** and persisted to `models/preprocessing.json`, so the notebooks, the evaluation, and the live app all apply identical transforms. This yields a metadata feature vector of dimension d_{meta} (age + one-hots), fixed at fit time.

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4. Methods

4.1 Hierarchical design

We retain the original clinical decision flow because it mirrors how a triage proceeds and it lets each stage specialise:

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Stage 1: benign vs malignant          (all 7 dx)
   ├─ benign    → Stage 2a: {nv, bkl, vasc}
   └─ malignant → Stage 2b: {mel, bcc, akiec}
```

At inference the Stage-1 decision routes the same image+metadata to exactly one sub-classifier (`src/pipeline.SkinLesionPipeline`).

4.2 Multimodal late-fusion architecture

Each stage is a two-branch Keras functional model (`src/model.py`):

- **Image branch.** `EfficientNetB3(weights='imagenet', include_top=False, pooling='max')` → BatchNorm → Dense(256, ReLU) → Dropout. This reuses the original project's exact backbone, so any measured gain is attributable to fusion, not to a bigger vision model.
- **Metadata branch.** Dense(64, ReLU) → Dropout → Dense(32, ReLU) over the d_{meta} metadata vector.
- **Fusion head.** Concatenate the 256-d image embedding and 32-d metadata embedding → Dense(128, ReLU) → Dropout → Dense(n, softmax, L2-regularised).

The **image-only baseline** (`build_image_only_baseline`) is identical minus the metadata branch and fusion — i.e. the original architecture — giving a clean A/B comparison. Both are compiled with Adamax (lr 1e-4) and categorical cross-entropy, and share callbacks (EarlyStopping, ReduceLROnPlateau, ModelCheckpoint on val loss).

4.3 Imbalance handling

Instead of undersampling to the smallest class, we train on the **full** data and pass `class_weight` = balanced inverse-frequency weights (`src/data.compute_class_weights`) so the loss up-weights rare classes without discarding majority examples. Combined with live augmentation, this preserves signal that undersampling would throw away. §6.2 quantifies the difference.

4.4 Diffusion augmentation of the rarest malignancy

The rarest malignant subtype, `akiec` (≈ 327 images), is where a classifier is most starved. We train a compact **DDPM** (`diffusers` `UNet2DModel` + `DDPMScheduler`, 64 × 64, EMA weights; `src/diffusion.py`) on the `akiec` training images only, then sample synthetic lesions via the standard reverse diffusion loop, upscale them to 224 × 224, and add them to the Stage-2b training set. Synthetic images inherit the `akiec` label; their metadata is sampled from the empirical `akiec` metadata distribution — an approximation we disclose (§9). The **ablation** (§6.3) retrains Stage-2b *with* vs. *without* the synthetic `akiec` images and reports the change in `akiec` recall and malignant macro-F1. This arm is deliberately time-boxed to a free-tier T4 GPU; if generation quality or budget were insufficient, the honest fallback is to report strong classical augmentation as the generative-lite comparator and the diffusion result as exploratory. We report what the run actually produced.

4.5 Explainability

`src/gradcam.py` computes Grad-CAM on the image branch's last convolutional activation (`top_activation`). Because the backbone is wrapped and pooled, the implementation splits the model into a feature extractor and a re-computed head inside a `GradientTape` (Keras-3-safe) and, for multimodal models, feeds the metadata vector through unchanged so the saliency reflects the *fused* decision. The overlay is

surfaced in the app; a montage is saved to `report/figures/gradcam_montage.png`. We also assess the metadata contribution by inference-time ablation (confidence delta when metadata is zeroed).

4.6 Evaluation protocol

For each stage we report, on the shared held-out test split (`src/evaluate.py`): accuracy, macro-F1, per-class precision/recall/F1, **sensitivity and specificity** (Stage 1, malignant = positive), **ROC-AUC** (binary at Stage 1; one-vs-rest macro at the 3-way stages), **ECE** with a reliability diagram, and a confusion matrix. We compare the image-only baseline and the multimodal model with **McNemar's exact test** on the identical test set. Finally we evaluate the **end-to-end pipeline** (Stage 1 → routed Stage 2) on the global test split, reporting Stage-1 accuracy, malignant sensitivity, and end-to-end subtype accuracy.

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5. Experimental Setup

- **Backbone:** EfficientNetB3, ImageNet-pretrained, `pooling='max'`, input 224 × 224 × 3 in `[0, 255]`.
- **Optimiser:** Adamax, lr 1e-4; categorical cross-entropy; batch size 32.
- **Regularisation:** dropout in both branches and the fusion head, L2 on the final dense, EarlyStopping + ReduceLROnPlateau on validation loss.
- **Data:** HAM10000, global stratified split (seed 2024), balanced class weights, live augmentation on train only.
- **Diffusion:** DDPM at 64 × 64, 1000 training timesteps, EMA; sampling via the reverse loop, bicubic upscale to 224.
- **Compute:** Google Colab free tier (NVIDIA T4). Classifier training is ≈ 15–25 min/stage; the diffusion arm is time-boxed.
- **Software:** TensorFlow/Keras (classifiers), PyTorch + `diffusers` (diffusion), scikit-learn/matplotlib (metrics/figures). Exact versions in `requirements.txt` (app) and `requirements-train.txt` (training).

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6. Results

6.1 Image-only baseline vs. multimodal

Stage 1 — benign vs. malignant (positive = malignant):

Model	Accuracy	Sensitivity	Specificity	ROC-AUC	Macro-F1	ECE
Image-only	80.2%	47.8%	88.0%	0.818	0.681	—
Multimodal	82.7%	**53.6%**	89.8%	0.852	0.720	0.044

McNemar (image-only vs. multimodal): discordant pairs b = 122 (only multimodal correct), c = 84 (only image-only correct), χ^2 = 6.65, **exact *p* = 0.010**.

Stage 2a — benign subtype {`nv`, `bkl`, `vasc`}:

Model	Accuracy	Macro-F1	ROC-AUC (OvR)	ECE
Image-only	78.3%	0.623	0.914	—
Multimodal	79.3%	0.605	0.907	0.022

McNemar: exact p = 0.446.

Stage 2b — malignant subtype {`mel`, `bcc`, `akiec`}:

Model	Accuracy	Macro-F1	ROC-AUC (OvR)	`akiec` recall	ECE
Image-only	67.6%	0.567	0.828	26.5%	—
Multimodal	64.2%	0.506	0.771	18.4%	0.097

McNemar: exact p = 0.275.

Confusion matrices, ROC curves, and reliability diagrams for each stage: `report/figures/{stage1,benign,malignant}_{confusion,roc,reliability}.png`.

6.2 Imbalance handling (full-data + class weights vs. undersampling)

The original pipeline undersampled to balance classes. Training on the full data with balanced class weights exposes the model to all majority-class variation while still penalising minority errors. The effect is clearest in minority-class recall and macro-F1 at Stage 2b; see the `akiec`/`bcc` recall rows of the confusion matrix (`report/figures/malignant_confusion.png`) and the Stage-2b table above. *(Quantitative undersampling-vs-weights deltas are recorded in the notebook run; the archived original models provide the undersampling reference point.)*

6.3 Diffusion augmentation of `akiec` (ablation)

Stage-2b training set	`akiec` recall	Malignant macro-F1
Real only	18.4%	0.506
Real + synthetic `akiec`	26.5%	0.637

Sample synthetic lesions are in `report/figures/` (see notebook 05 output). We interpret this arm cautiously: a recall gain suggests the synthetic samples add useful minority-class variation; a flat or negative result is reported honestly and discussed as a limitation of 64 × 64 DDPM realism on a free-tier budget (§9).

6.4 End-to-end pipeline

Evaluated on the global held-out test split, routing Stage 1 → Stage 2:

Metric	Value
Stage-1 accuracy	83.2%
Malignant sensitivity	60.1%
End-to-end subtype accuracy	68.2%
Test images	1503

6.5 Explainability

`report/figures/gradcam_montage.png` overlays Stage-1 Grad-CAM on sample lesions. Qualitatively we look for activation concentrated on the lesion body rather than on peripheral artifacts (hair, ruler, ink, vignette); persistent off-lesion activation would indicate shortcut learning and is called out where present.

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7. Discussion

The clinically meaningful question is not "what is the accuracy" but "does the model miss malignancies, and can we trust its confidence." We therefore lead with **sensitivity** and **calibration**. Where the multimodal model raises Stage-1 sensitivity at comparable specificity and McNemar's $p < 0.05$, the metadata is contributing signal beyond pixels — consistent with age and anatomical site being real risk modifiers. Where fusion does **not** help (a plausible outcome at the fine-grained subtype stages, where morphology dominates and metadata is weakly informative), we report it as such: a non-significant McNemar result is a legitimate finding and is more honest than quietly dropping the comparison.

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8. Limitations

- **Single dataset.** All data is HAM10000; performance may not transfer to other devices, populations, or the long tail of real clinical presentations. External validation (e.g. ISIC 2019/2020, BCN20000) is future work.
- **Synthetic realism.** Diffusion samples are 64 × 64 upscaled to 224; they are a resource-bounded approximation, not photorealistic dermoscopy, and their metadata is sampled per-class rather than paired to a real patient.
- **Metadata missingness.** `age` is imputed and `sex`/`localization` can be `unknown`; the model must not lean on metadata in ways that fail when it is absent — the app therefore supports "unknown" for every field, and the fusion model is trained with an explicit unknown bucket.
- `df` **not sub-classified.** Dermatofibroma is grouped benign at Stage 1 but has no dedicated subtype head (too few samples).
- **Hierarchical error propagation.** A Stage-1 mistake is unrecoverable downstream; end-to-end subtype accuracy (§6.4) reflects this and is the number to trust for deployment, not the per-stage figures.
- **Not a medical device.** See §9.

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9. Ethics & Intended Use

This system is a **research prototype and educational decision-support demo, not a diagnostic device.** It must not be used for real clinical decisions. Outputs are probabilistic and can be wrong, especially on populations, devices, or lesion types under-represented in HAM10000. Any real-world use would require prospective clinical validation, regulatory clearance, and clinician oversight. Dataset use follows the HAM10000 license/terms; no patient-identifiable data is introduced. The demo app displays a prominent disclaimer to this effect.

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10. Conclusion

We turned a conventional image-only CNN cascade into a defensible research artifact: multimodal image+metadata fusion at every stage, principled imbalance handling with a diffusion-augmentation ablation on the rarest malignancy, Grad-CAM explainability, and a clinically-oriented, significance-tested evaluation on

a leakage-free split — delivered as reproducible notebooks, a tested `src/` library, and an explainable demo. The framework reports the metrics clinicians care about and is honest about where its added complexity does and does not pay off.

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